hold equities, but this agent wishes to have extremely smooth consumption paths over time. Risk-free rates need to be very high to induce them not to smooth consumption. It turns out that the question, "why are equity returns so high?" is the flipside of the question, "why are risk-free rates so low?" If we're *over* compensated for investing in equities, we're *under*compensated for investing in bonds.

A class of models with *time-varying risk aversion* can explain the equity premium puzzle. Let us illustrate how they work by looking at the model of *habit utility*. We have come across these utility functions before, in chapter 2, where they were used to construct utility of wealth relative to some level, rather than using the utility of wealth itself. In habit utility, the utility of the representative agent depends on consumption relative to past consumption or a *habit level*. A canonical model is by Campbell and Cochrane (1999).

Consider a young college graduate who is accustomed to sleeping on sofas, sharing apartments, and driving second-hand cars. She gets a job as a barista. Then she suddenly gets laid off. This is, of course, a bad time, but it does not hurt that much because she is already used to her frugal lifestyle. Her consumption has not declined by much relative to her habit. Now suppose that upon graduation she lands a high-flying job (on Wall Street, I guess). Her consumption increases because her income is now very high. She moves into her own place with a wonderful view, buys a brand new car, and starts a wardrobe of designer clothing. Her consumption habit has increased. Now when she is suddenly laid off, this is a *very* bad time. She actually might have more money from some savings than as a barista, but her habitual consumption level is very high, and so the drop in her consumption relative to her habit hurts a lot. Her *marginal utility* is very high because she is used to having high consumption and has to go back to low consumption. In habit models, it is not the consumption decrease per se that matters; what matters instead is the consumption decrease relative to habit.

Habit models allow *local* risk aversion to be very high. When recessions hit, risk aversion can shoot up to very high levels—the risk aversions of 20 or even 100 required by equation (8.1)—but the increase is temporary. In economists' jargon, marginal utility is very high during recessions as consumption approaches habit; the agent really values that extra $1, and the curvature of the utility function is very steep. Although consumption itself barely falls during recessions, the small reductions in consumption bring agents perilously close to their habits. Thus, agents become very risk averse during bad times, generating high equity risk premiums. Equity prices fall in recessions to generate high future returns.

In good times, consumption is far above habit. Risk aversion is low, and equity premiums are small. In these good times, marginal utility is low, and the representative agent's utility function is very flat. During booms, equities are expensive, and there is not much room for future appreciation; hence, future expected returns are low. Since expected returns are high during bad times and low during good times, another benefit of time-varying risk aversion models is that they match how equity returns can be predicted over time, a subject that we cover in